x

$$mu_x - mv_\infty = \int G\frac{Mm}{r^2}cos\varphi dt = \int G\frac{Mm}{r^2\dot{\varphi}}cos\varphi d\varphi. \quad (18)$$

$$r^2\dot{\varphi} = v_\infty b. \quad (19)$$

$$mu_x - mv_\infty = G\frac{Mm}{v_\infty b}\int_0^\theta cos\varphi d\varphi = G\frac{Mm}{v_\infty b}sin\theta. \quad (20)$$

y

$$mu_y - m\cdot 0 = G\frac{Mm}{v_\infty b}\int_0^\theta sin\varphi d\varphi = G\frac{Mm}{v_\infty b}(1-cos\theta). \quad (21)$$

$$z = \frac{GM}{v_\infty^2 b}, \quad (22)$$

$$u_x = (1+zsin\theta)v_\infty, \quad (23)$$
$$u_y = z(1-cos\theta)v_\infty. \quad (24)$$

(23) (24) (17)

$$1 = (1+zsin\theta)^2 + z^2(1-cos\theta)^2 - 2z\frac{b}{R}. \quad (25)$$

θ

$$\theta = \arcsin\frac{\frac{b}{R}-\frac{GM}{v_\infty^2 b}}{\sqrt{1+\left(\frac{GM}{v_\infty^2 b}\right)^2}} + \arcsin\frac{\frac{GM}{v_\infty^2 b}}{\sqrt{1+\left(\frac{GM}{v_\infty^2 b}\right)^2}}, \quad (26)$$

$$\theta = 2\arctan\frac{1-\sqrt{1+2\frac{GM}{v_\infty^2 b}\frac{b}{R}-\frac{b^2}{R^2}}}{\frac{b}{R}-2\frac{GM}{v_\infty^2 b}} = 0.789\ rad = 45.2°. \quad (27)$$

각도 θ는 우주선이 먼지 구름 속을 이동할 때 궤적의 전체 편향각과 마찬가지로, 극좌표로 쓴 에너지 보존 법칙과 각운동량 보존 법칙을 결합한 방정식을 적분하여 구할 수 있다. 적분이 상당히 복잡하므로 그 식은 여기에서 제시하지 않는다.

2.2 **[2.0점]**

$r < R$.

r ,

$$F(r) = -G\frac{\rho_4\cdot\frac{4}{3}\pi r^3}{r^2}m = -\frac{4}{3}\pi G\rho_4 mr, \quad (28)$$

$$U(r) = -\int F(r)dr = \frac{2}{3}\pi G\rho_4 mr^2 + C = G\frac{Mm}{2R^3}r^2 + C. \quad (29)$$

C
$r = R$

$$G\frac{Mm}{2R^3}R^2 + C = -G\frac{Mm}{R}, \quad (30)$$

$r < R$

$$U(r) = \frac{GMm}{2R^3}r^2 - \frac{3GMm}{2R}. \quad (31)$$

.

$$\frac{mv_\infty^2}{2} = \frac{mv_0^2}{2} + \frac{GMm}{2R^3}r_{min}^2 - \frac{3GMm}{2R}, \quad (32)$$
$$v_0 r_{min} = v_\infty b, \quad (33)$$